

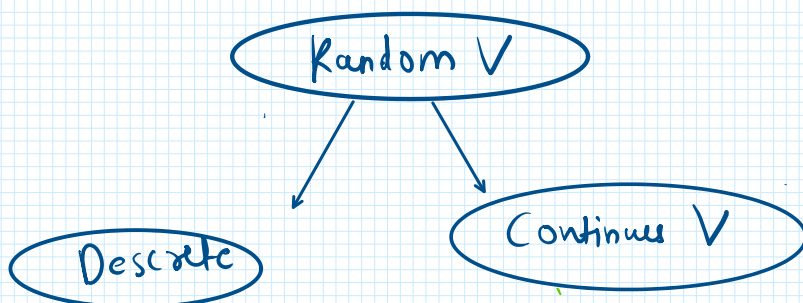
① Random Variable $\Rightarrow$ It is a way to give numbers to the outcome of a Random Event

$\{g \Rightarrow \bigcirc \rightarrow$

$$X = \begin{cases} 1, & \text{if } \underline{\text{Head}} \\ 0, & \text{if Tail} \end{cases}$$

Represent:

x, y, z



→ count.

→ Dice $\rightarrow \{1, 2, 3, 4, 5, 6\}$

→  → Output Design


$$\rightarrow [0 \text{ --- } 1]$$

$0 \rightarrow 1$ $(2) \rightarrow \text{Dice}$

Sample Space:

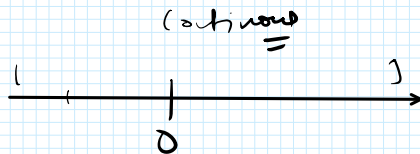
$$\{(1,1) (1,2) \dots (1,6)\}$$

(3,3)
(6,1) - - - - - (6,6)

Event $\rightarrow$ Two Dice = 7

$$X=7 = \{(1,6) (3,4) (4,3) (5,2) (6,1)\}$$

$$\frac{5}{36} \rightarrow \frac{1}{6} = \text{PMF}$$



★ PMF $\Rightarrow$ (Probability mass function)

$\Rightarrow$ prob. of each possible value of a Random V.

Random Variable

0.

$$X = \begin{cases} 1, & \text{if } H \\ 0, & \text{if } T \end{cases}$$

Possible outcomes $\Rightarrow 0, 1$

Prob of Head $\Rightarrow 1/2$

" " Tail $= 1/2$

PMF $P(X=\underline{x}) \Rightarrow P(X=7)$

$$P(X=1) = 1/2 = 0.5$$

$$P(X=0) \Rightarrow 1/2 = 0.5$$

★ NOTE

Total outcome Sum = 1.

$$\sum_{i=0}^n P(X=x_i) = 1$$

$$n=2$$

$$0 \leq P(X=x) \leq 1$$

$$0, 1, \dots, n$$

1) A random variable takes the values 1, 2, 3, 4, and 5. Table 8.5.1 gives the PMF $P(X=x)$ of X .

X	1	2	3	4	5
$P(X=x)$	0	$\frac{t}{2}$	$4t^2$	$\frac{3t}{2}$	

Table 8.5.1: Probability mass function

Find the probability of $X = 5$ ($P(X=5)$).

$$P(X=5) \Rightarrow \frac{3t}{2} \Rightarrow \underline{\underline{1}}$$

$$\text{Ans} \Rightarrow 0 + t + \frac{t}{2} + 4t^2 + \frac{3t}{2} = 1$$

$$\Rightarrow \frac{3t}{2} + \frac{t}{2} = \frac{4t}{2} = 2t = t + t + 0 + 4t^2$$

$$3t + 4t^2 = 1$$

$$\Rightarrow 4t^2 + 3t - 1 = 0$$

$$4t^2 + 4t - t - 1 = 0 \quad \Rightarrow +4t - t$$

$$(4t - 1)(t + 1) = 0$$

$$t = \frac{1}{4}, t = -1$$

$$\frac{1}{4} \checkmark$$

$$\frac{3}{8} \checkmark$$

$$\frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$$

Q →

3) Probability mass function of a random variable is given by:

$$P(X=x) = \begin{cases} k & \text{if } x=0 \\ 0.4 + 2k & \text{if } x=1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k ✓

$$P(X=0) = k$$

$$P(X=1) = 0.4 + 2k$$

$$\Rightarrow k + 0.4 + 2k = 1$$

$$3k + 0.4 = 1$$

$$k = \frac{0.6}{3} = 0.2 \checkmark$$

H.W
Q = 1 ✓

6) The probability mass function of a discrete random variable X is given by

$$P(X=x) = k({}^2C_x \cdot {}^3C_{3-x}) ; \text{ for } x = 0, 1, 2$$

Find the value of k

Q = 2 ✓

5) Let X be the number of candies present in a box. We have the following information:

There are at most three candies in the box.

The probability of having 2 candies in the box is the same as the probability of having one candy.

The probability of having no candy in the box is the same as the probability of having 3 candies.

The probability of having 1 or 2 candies is half of the probability of having 0 or 3 candies.

What will be the PMF of X ?

Q = 3 ✓

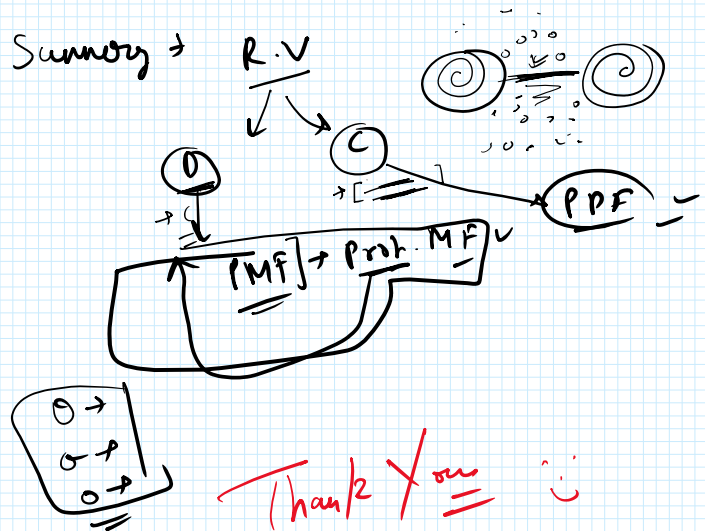
There are five multiple choice questions asked in an exam. There is a chance of 70% that Shreya will solve a question correctly. Let X be the random variable that represents the number of questions she solves correctly. Which of the following is the probability mass function of X ? (Assume that the correctness of answers to different questions are independent)

$$P(X=x) = \begin{cases} 0.00243 & \text{for } x=0 \\ 0.02835 & \text{for } x=1 \\ 0.1323 & \text{for } x=2 \\ 0.3087 & \text{for } x=3 \\ 0.36015 & \text{for } x=4 \end{cases}$$

$$P(X=x) = \begin{cases} 0.00243 & \text{for } x=0 \\ 0.02835 & \text{for } x=1 \\ 0.1323 & \text{for } x=2 \\ 0.3087 & \text{for } x=3 \\ 0.36015 & \text{for } x=4 \\ 0.16807 & \text{for } x=5 \end{cases}$$

$$P(X=x) = \begin{cases} 0.02835 & \text{for } x=1 \\ 0.1323 & \text{for } x=2 \\ 0.3087 & \text{for } x=3 \\ 0.36015 & \text{for } x=4 \\ 0.16807 & \text{for } x=5 \end{cases}$$

$$P(X=x) = \begin{cases} 0 & \text{for } x=0 \\ 0.03078 & \text{for } x=1 \\ 0.1323 & \text{for } x=2 \\ 0.3087 & \text{for } x=3 \\ 0.36015 & \text{for } x=4 \\ 0.16807 & \text{for } x=5 \end{cases}$$



Thank You ☺



27 →
21

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★ Revision → But ✓